ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics practice exam 3.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable symbols.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 B ✓ 1.1.4 A ✓ 1.1.5 D ✓ 1.1.6 C ✓ 1.1.7 B ✓ 1.1.8 C ✓ 1.1.9 A ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 J ✓ 1.3.3 F ✓ 1.3.4 E ✓ 1.3.5 H ✓ 1.3.6 A ✓ 1.3.7 G ✓ 1.3.8 B ✓ 1.3.9 D ✓ 1.3.10 I ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Protons – positive, in the nucleus ✓; neutrons – no charge, in the nucleus ✓; electrons – negative, orbiting around the nucleus in shells ✓ (3)

2.2 Metals have many free (loosely held) electrons in their outer shells ✓ that can move easily from atom to atom when a voltage is applied ✓. (2)

2.3 (a) 4,7 kΩ ✓ (b) 25 mA ✓ (c) 2,2 × 10^{-6} F ✓ (d) 3 000 V ✓ (4)

2.4 Any FIVE ✓✓✓✓✓: no food or drink; wear safety goggles when soldering or cutting; switch off before working on a circuit; never work with wet hands; check cables and tools for damage; return the soldering iron to its stand; tie back long hair / no loose clothing; know where the fire extinguisher and first-aid kit are; report injuries and faults. (5)

[14]

QUESTION 3

3.1.1 $R_T = 100 + 220 + 680 = 1\,000\ \Omega$ ✓ (1)

3.1.2 $I = V \div R$ ✓ $= 12 \div 1\,000 = 0{,}012\ A\ (12\ mA)$ ✓ (2)

3.1.3 $V = I \times R$ ✓ $= 0{,}012 \times 680 = 8{,}16\ V$ ✓ (2)

3.2 $R_T = (6 \times 3) \div (6 + 3)$ ✓ $= 2\ \Omega$ ✓; $I = 12 \div 2 = 6\ A$ ✓; $I_{6\Omega} = 12 \div 6 = 2\ A$ ✓ (4)

3.3.1 $R_P = (6 \times 12) \div (6 + 12) = 4\ \Omega$ ✓; $R_T = 4 + 4 = 8\ \Omega$ ✓ (2)

3.3.2 $I = 24 \div 8 = 3\ A$ ✓ (1)

3.3.3 $V_P = 3 \times 4 = 12\ V$ ✓ (1)

3.3.4 $I_{12\Omega} = 12 \div 12 = 1\ A$ ✓ (1)

3.4.1 $W = P \times t = 0{,}06\ kW \times 5 \times 20$ ✓ $= 6\ kWh$ ✓; $cost = 6 \times R2{,}50 = R15{,}00$ ✓ (3)

3.4.2 $Q = I^2Rt$ ✓ $= 2^2 \times 10 \times 60$ ✓ $= 2\ 400\ J$ ✓ (3)

[20]

QUESTION 4

4.1 Yellow 4, violet 7 ✓, orange × 1 000 ✓ = 47 000 Ω (47 kΩ); gold = ±5% ✓ (3)

4.2 (a) $C_T = 10 + 10 = 20\ \mu F$ ✓ (b) $1/C_T = 1/10 + 1/10$ ✓; $C_T = 5\ \mu F$ ✓ (3)

4.3 Doping is adding a small amount of another element (impurity) to pure silicon to change how it conducts ✓. N-type: add an element with five outer electrons (e.g. phosphorus / arsenic) ✓, giving extra free electrons ✓. P-type: add an element with three outer electrons (e.g. boron / gallium), giving holes ✓. (4)

4.4 (a) triangle pointing to a bar ✓ (b) diode symbol with two arrows pointing out ✓ (c) one straight and one curved (or open and filled) plate, with + marked ✓ (d) a break in the line with a hinged contact ✓ (4)

4.5 Without it the LED would draw too much current and burn out; the resistor limits the current ✓. $R = (V_s - V_{LED}) \div I$ ✓ $= (9 - 2) \div 0{,}02$ ✓ $= 350\ \Omega$ ✓ (4)

[18]

QUESTION 5

5.1 (a) 45 = 32 + 8 + 4 + 1 ✓ → 101101 ✓ (b) 128 + 64 + 16 + 4 + 2 ✓ = 214 ✓ (c) 214 ÷ 16 = 13 remainder 6 ✓ → D6 ✓ (6)

5.2 OR: 00→0, 01→1, 10→1, 11→1 ✓✓. NAND: 00→1, 01→1, 10→1, 11→0 ✓✓ (4)

5.3.1 $Y = A \cdot B + C$ ✓✓ (2)

5.3.2 8 rows (2^3) ✓ (1)

5.3.3 1 (C alone makes Y = 1) ✓ (1)

5.4 Analogue: changes smoothly and can have any value in a range ✓, e.g. a temperature sensor voltage, a microphone signal ✓. Digital: has only two levels, 0 and 1 (OFF and ON) ✓, e.g. a push button, the output of a logic gate ✓. (4)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE